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Palaeomagnetism and magnetic fabric in the Freetown Complex, Sierra Leone

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SUMMARY

About six separately orientated cores were collected at each of 14 sites distributed throughout the arcuate, west-dipping, 6 km thick, Freetown layered igneous complex. Alternating field and thermal demagnetization both isolate a stable component of remanent magnetism which corresponds to a palaeomagnetic south pole from 13 sites (nine reverse, four normal polarity) at 82.9°S, +32.7°E ($\alpha_{95} = 5.6°$). This is indistinguishable from that reported in 1971 based on alternating field demagnetization of cores from 10 orientated hand samples.

The difference between the Freetown pole (age: 193 ± 3 Ma) and other mid-Jurassic poles from West Africa could be due to its greater age. The difference between the whole West African Jurassic pole group and the Karoo pole from southern Africa, however, suggests moderate ($\sim 10°$) differential rotation of West Africa relative to the Kaapvaal craton.

A prevalent magnetic foliation fabric coincides generally with the petrological layering, as might be expected, but a ubiquitous magnetic lineation is predominantly down-dip. This is compatible with a down-dip pyroxene lineation reported to be present in some field outcrops, and interpreted in terms of late-stage deformation during the slow crystallization and cooling of the large igneous body. However, a fold test shows that the igneous layering had already achieved its present attitude before the Complex cooled to ~ 570 °C (the maximum blocking temperature of the characteristic remanence).

Key words: igneous petrology, magnetic fabric, palaeomagnetism.

INTRODUCTION

Briden *et al.* (1971) reported results of a palaeomagnetic study of orientated samples collected from the Freetown Complex in which, after alternating field (AF) cleaning, consistent vectors were observed, and a pole was calculated. Beckinsale *et al.* (1977) published a Rb–Sr whole-rock isochron age of 193 ± 3 Ma from late-stage acid veins in the Complex. Thus, this well-dated intrusive body was emplaced penecontemporaneously with the extensive Karoo lavas in South Africa (183 ± 2 Ma, Duncan *et al.* 1997), affording an opportunity for comparison of roughly synchronous poles determined from widely separated places within a single plate. Unfortunately, the original Freetown pole was based on results from only 10 individual hand samples, although these were collected at various locations throughout the exposed complex. Given the good age determination, however, upgrading of the pole by more extensive sampling was justified, and the results are reported here.

GEOLOGY

The study of the Freetown Complex has a long history, the primary foundation being Wells (1962). Petrological research since has added much to the knowledge of mineral chemistry and textural relations of various phases (Beckinsale *et al.* 1977; Bowles 1976, 1977, 1978; Wells & Bowles 1981; Umeji 1983; Chalokwu *et al.* 1995; Chalokwu & Seney 1995). The Complex is an impressive layered igneous body over 6 km thick, the eastern segment of which is exposed on land in a mountainous arc some 60 km long, forming the peninsula south of Freetown

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(see Fig. 1a). The intrusion appears funnel-shaped, with the dip of layering increasing from about 15° at the base to >40° on the coast (Fig. 1a).

The offshore geophysical work of Krause (1963) was interpreted to indicate that the structure formed a circular cone of radius ~30 km. However, more recent and extensive offshore geophysical work (Mgbatogu *et al.* 1988) belied this simple shape. Modelling based on their offshore magnetic and gravity survey together with the onshore data led Mgbatogu *et al.* (1988) to conclude that '…it is an elongated feature, some 150 km in length with a NW–SE trend. Gravity modelling indicates that the intrusion has a maximum width of about 50 km and reaches depths exceeding 17 km south-west of Freetown….' The exposed Freetown Complex is thus but part of a very large igneous intrusion.

According to Wells (1962), the exposed rocks comprise four successive rhythmically layered sequences or zones, each grading from predominantly troctolite at the base up to anorthositic gabbro at the top. No systematic change in the composition of the major phases throughout the complex was detected by Wells (1962), who drew particular attention to the apparent absence of cryptic variation. The origin of the Complex is attributed to a multiplicity of intrusion pulses with relatively minor fractionation, and/or crystallization attended by high volatile concentrations. According to Umeji (1983), the latter is inferred to have fostered high fO_2, which instigated the early crystallization of Fe–Ti oxides. This, Umeji suggested, caused depletion of the magma in iron, thereby minimizing the possibility of variation in Mg/Fe in the silicates. More recently, however, Chalokwu *et al.* (1995) and Chalokwu & Seney (1995) have reported clear evidence of cryptic variation based on their detailed microprobe study. They conclude that the pattern of variation 'suggests continual leakage of fresh magma into the chamber, followed by oscillatory spikes in the rhythmically layered subzone of Zone 3, where major influxes of new magma occurred' (Chalokwu *et al.* 1995; p. 230).

The acid veins which were dated by Beckinsale *et al.* (1977) are deemed to represent volatile-enriched acidic residues from crystallization of the basic phases. The late growth of large, elongate pyroxene oikocrysts, dipping more steeply than the original layering, was documented by Wells (1962) and Wells & Bowles (1981). This must reflect structural adjustments accompanying the intrusion and progressive crystallization of the body.

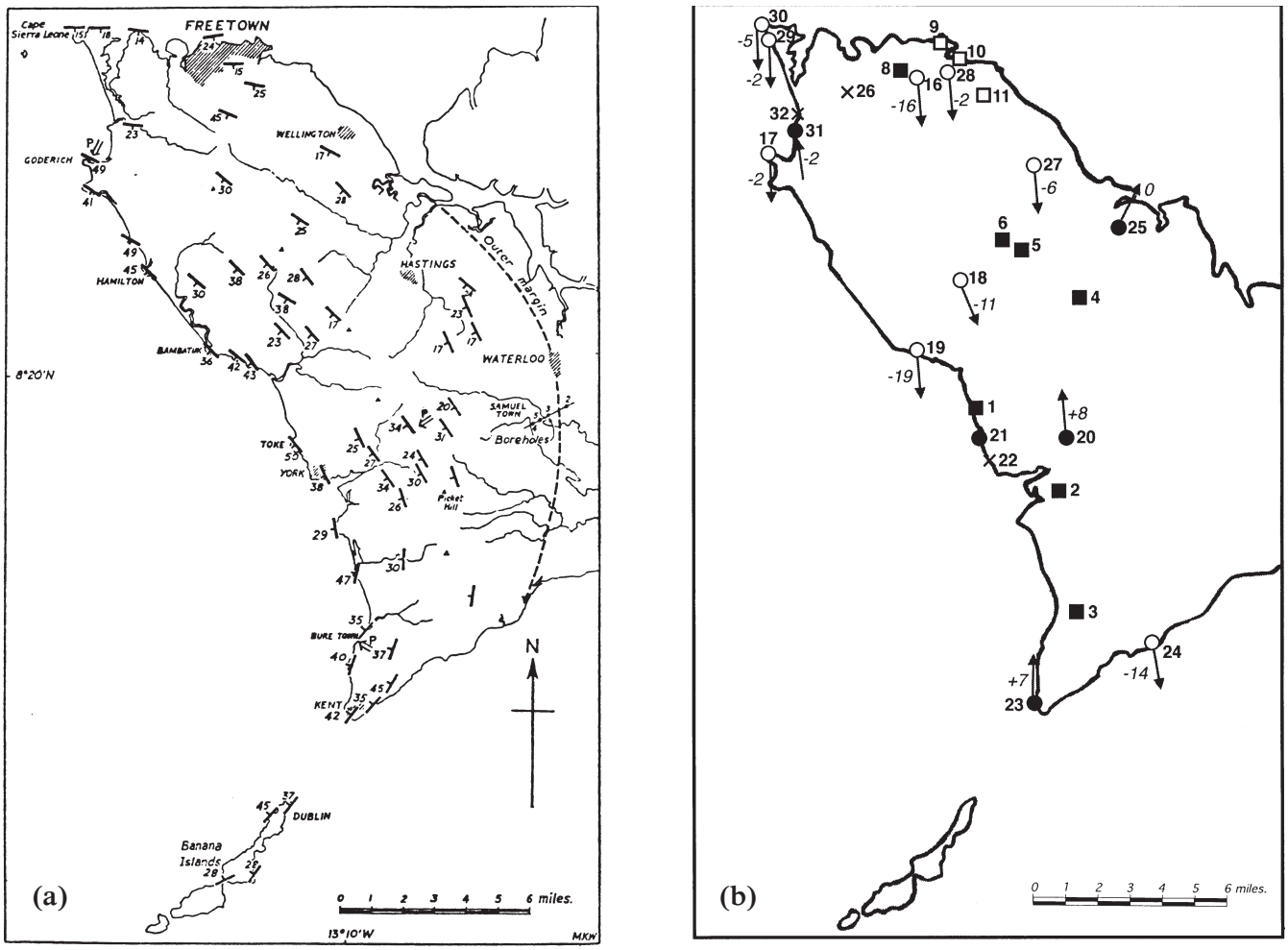


Figure 1. (a) Outline of map of Freetown Complex showing strike and dip of layering. Double arrows signify pyroxene lineation. Adapted from Plate 1 in Wells (1962). (b) Numbered palaeomagnetic sampling sites: closed circles, normal polarity; open circles, reverse polarity; crosses: younger dykes. Samples of Briden *et al.* (1971): closed squares, normal polarity; open squares, reverse polarity. Mean-site palaeomagnetic vectors from this study: the arrow is in the direction of the declination *D*, and the inclination *I* is marked alongside.

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FE–TI OXIDES AND MAGNETIC PROPERTIES

The most prominent magnetic Fe–Ti phases present consist of large (up to ~mm), roughly equant, granular aggregates of magnetite-ilmenite (Figs 2a and c). These vary irregularly in abundance, through the various rock types, and some discrete oxide-rich layers are reported (Wells 1962). The invariable association of magnetite with ilmenite in these individual granular grains must have resulted from crystallization of an ulvospinel-rich magnetite, followed by high-temperature oxidation and complete oxy-exsolution of ilmenite. The composition of the coexisting cubic and rhombohedral phases indicates equilibration down to temperatures of ~600 °C and a log fO_2 of −19.0 (Bowles 1977; Chalokwu *et al.* 1995). The magnetite appears homogeneous—with only minor ceylonite (?) exsolution, consistent with its reported elemental composition (Bowles 1977). The grains are clearly of a multidomain size. They are the source of high susceptibility, and carriers of strong isothermal remanence (IRM). The ilmenite, too, is homogeneous, save for occasional magnetite lamellae parallel to (0001) (as confirmed by the application of magnetic colloid). These—a product of reductio-exsolution (which is the converse of ilmenite oxy-solution from magnetite)—indicate that the ambient fO_2 during the cooling of these Fe–Ti–O phases became too low to be compatible with the initial Fe_2O_3 content of the ilmenites. These magnetite lamellae, though rare, are elongate and probably of a size appropriate for pseudo-single-domain magnetic properties, and so could carry a stable remanence.

As confirmed by the study of polished thin sections and application of magnetic colloid, however, most fine-grained magnetite is present in the form of abundant schiller-intergrowth rodlets in the diallagic clinopyroxene (Figs 2e and f; see also Wells 1962; p. 44–46). There is also exsolved orthopyroxene in these augites, the form and orientation of the lamellae in some grains being parallel to both (001) and (010), the former habit testifying to an origin as monoclinic pigeonite lamellae. The magnetite rodlets occur most commonly (but not exclusively) along the interface with the orthopyroxene lamellae (Wells 1962; Wells & Bowles 1981). The augite grains are rarely without these magnetite-rodlet schiller intergrowths, and given their size and shape anisotropy, these are almost certainly the predominant carrier of the stable characteristic remanent magnetism. Petrologically, the origin of such rodlets is not well understood, but is probably related to a high-temperature oxidation (?) reaction of Fe^{2+}, whereby magnetite 'exsolves' from the pyroxene host. Their growth would have occurred at high temperatures, well above the magnetite Curie point. In some samples, the plagioclase grains are distinctly clouded, with fine opaque rods (too small for identification) which, if magnetic, would also contribute to stable remanence. Thus, from microscopic study of the magnetic phases in these rocks, it is concluded that the characteristic remanence is most probably a TRM.

SAMPLING AND MEASUREMENT

On average, six cores were drilled in the field and orientated with Sun and magnetic compass at each of 14 sites distributed throughout the mainland part of the Complex (Fig. 1a). In addition, three younger dolerite dykes cutting Complex rocks were sampled. Following cutting of these cores, the palaeomagnetism of at least one specimen from each sample was measured in the Palaeomagnetic Laboratory at Oxford University. In reconnaissance, both thermal and AF stepwise demagnetization were applied, the latter proving more successful in isolating a characteristic remanence. Stepwise AF demagnetization up to 100 mT was then applied to all remaining specimens, and the results analysed by principal component analysis using the techniques of Kent *et al.* (1983). Site-mean directions and corresponding virtual geomagnetic poles (VGPs) were calculated from specimens in which the characteristic remanence (ChRM) was isolated; the results are summarized in Table 1. In addition, at Princeton the anisotropy of magnetic susceptibility (AMS) was measured for two specimens from every sample on a Sapphire Instruments SI2 susceptibility bridge linked to an IBM PC. These results are presented in Table 2.

PALAEOMAGNETISM

A very strong soft component present in most samples is believed to be IRM, but AF demagnetization usually revealed a clear characteristic remanence (Fig. 3). As can be seen in Table 1, samples at 13 of the 14 sites in the Complex gave reasonably consistent results, and site means could be calculated. Of these, nine sites carry reverse polarity ChRMs and four normal, the directions being virtually antipodal (Fig. 4). Incorporating the data of Briden *et al.* (1971), the distribution of the normal and reverse sites is as shown in Fig. 1(b). Towards the north end of the Peninsula, the polarities are predominantly reverse, whereas in the centre, and to the south, they are mostly normal. Reverse polarities at sites 19 and 24 appear to bracket the normal polarity sites, possibly suggesting that the top and bottom of the exposed complex are reversely magnetized, whereas the middle is normal. This would imply that, during cooling and magnetization of the complex, a polarity transition from reverse to normal occurred. There is no clear boundary between the two polarity zones, however, and in the north especially, where reverse polarity is prevalent, there is no correlation with the petrological zones.

That a field reversal occurred during the slow cooling and magnetization of the Complex rocks doubtless contributes to the unevenness of the results. Samples from seven of these sites [16, 17, 19, 27, 28, 30 (all R) and 23(N)] exhibit relatively simple results: at least four of the six samples from each site are similar, and the site means had 95 per cent confidence limits (α_{95}) of less than 10°. At the remaining sites, directional scatter (and in one case the occurrence of both polarities) required judgement as to the primary polarity; at these sites the α_{95}s were large (18° to 27°, corresponding to Fisher precision estimates of 20–12). It is inevitable that the rocks in some parts of the Complex acquired their magnetization while the polarity reversal was occurring; thus, depending upon the blocking temperature spectrum of individual samples, a complex TRM mixture may have resulted. The mean (reverse) direction of these 13 sites ($D = 174.8°$, $I = -6.9°$, $\alpha_{95} = 6.2$) yields a (south) palaeomagnetic pole at 83.0°S, 34.4°E, $\alpha_{95} = 6.3°$. Remarkably, this is almost identical to the pole originally reported by Briden *et al.* in 1971, at 81°S, 45°E. The presence of both polarities in the slowly cooled Complex rocks suggests that geomagnetic secular variation has been averaged and the calculated pole is a legitimate 'palaeomagnetic' pole. Attempts to explain the TRM origin of the ChRM by modified

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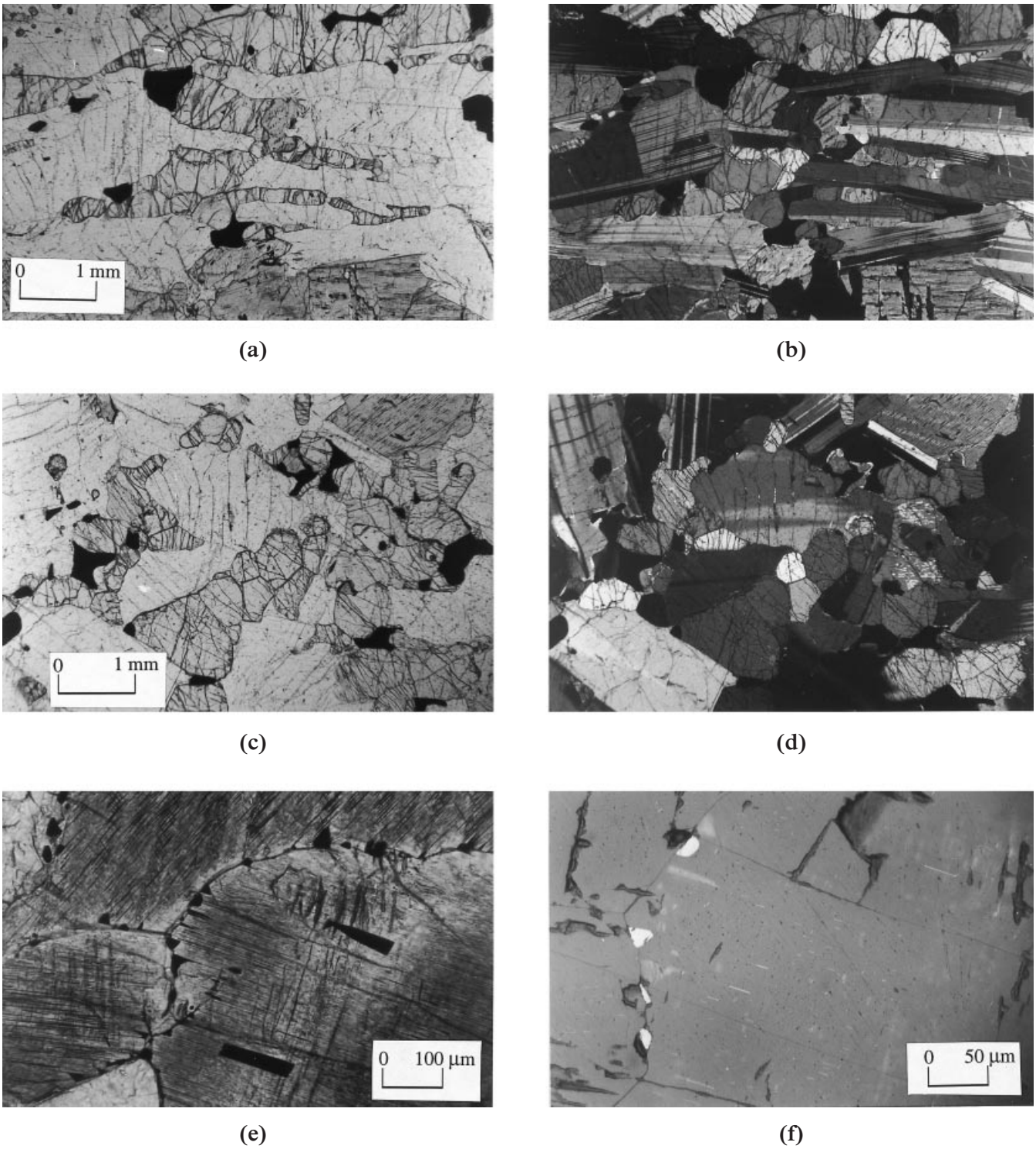


Figure 2. Photomicrographs to illustrate the contrast in mineral fabric in thin sections cut perpendicular (a, b, plane light and crossed polarizers) and parallel (c, d, plane light and crossed polarizers) to the magnetic foliation of sample 25-2. Clear grains in (a) and (c) are plagioclase, opaques are titaniferous magnetite, and irregularly fractured grains are olivine. Grains of augite with Schiller texture are at the bottom of (a) and top right of (c). (e, f) Schiller texture in pyroxene in sample 23-5, in plane (e) and reflected (f) light. Note the needle-like structure and diverse orientation of opaque rodlets, at least some of which are magnetite, and larger titaniferous magnetite grains between pyroxenes. Scale bars as indicated.

Thellier–Thellier experiments using the routine developed by McClelland & Briden (1996) were frustrated by the development at modest heating temperatures of magnetic alteration with a wide range of blocking temperatures.

The remanence directions in samples from the three younger dykes were scattered, and no significant mean direction could be calculated; the vectors were nevertheless of consistent polarity at each site, as indicated in Table 1.

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Table 1. Palaeomagnetic results.

Site	N/n	D°	I°	k	$\alpha_{95}{}^\circ$	Lat°N	Long°E	$\lambda\rho^\circ$N
16	6/5	170.9	−16.4	104	7.5	−81.0	77.3	8.4
17	6/6	176.1	−1.8	82	7.4	−81.6	14.6	0.9
18	6/5	154.3	−11.4	17	19.0	−64.4	72.9	5.7
19	6/6	173.9	−18.8	74	7.9	−83.8	90.4	9.7
20	7/6	352.0	+8.4	15	18.0	+81.1	230.8	4.2
21	No consistent results; polarity N?							
22 (dyke)	No consistent results; polarity N							
23	6/6	4.1	+6.5	344	3.6	+81.0	120.8	3.3
24	6/3	171.2	−13.3	47	18.1	−81.2	70.3	7.0
25	6/4	19.8	−0.3	12	27.4	+68.5	99.2	−0.2
26 (dyke)	No consistent results; polarity R							
27	6/6	171.1	−6.2	59	8.0	−79.7	46.8	3.1
28	6/6	173.4	−2.1	40	9.6	−80.2	29.0	1.1
29	6/4	176.8	+0.4	20	21.1	−80.8	7.2	−0.2
30	7/7	177.7	−4.7	91	5.8	−83.5	7.6	2.4
31	6/3	350.3	−1.9	670	4.8	+76.6	213.3	−1.0
32 (dyke)	No consistent results; polarity R							
Average of 9 Reversed Sites (16, 17, 18,19, 24, 27, 28, 29, 30)								
	9	171.8	−8.4	68	6.3	−80.9	51.0	
Average of 4 Normal Sites (20, 23,25, 31)								
	4	1.5	+3.2	31	16.7	+83.2	154.1	
Thirteen total sites								
	13	174.8	−6.9	46	6.2	−83.0	34.4	
	13*			56.5	5.6	−82.9	32.7	

N/n = samples measured/used; k and α_{95} Fisher statistical parameters
$\lambda\rho^\circ$N = site paleolatitude
* mean of site mean poles

Table 2. Magnetic susceptibility anisotropy results.

Site	n	$\chi \times 10^{-6}$ SI	K_1/K_3	K_1/K_2	FOL	LIN	Min (K_3) Dec°	Min (K_3) Inc°	Min (K_3) $\alpha_{95}{}^\circ$	Max (K_1) Dec°	Max (K_1) Inc°	Max (K_1) $\alpha_{95}{}^\circ$
16	12	5600	1.17	1.04	13.8	3.7	47.2	80.4	2.9	197.9	8.7	11.3
17	12	6400	1.15	1.06	11.1	5.9	22.8	41.0	28.7*	235.7	43.6	23.4*
18	12	6500	1.09	1.03	6.7	2.9	49.9	65.0	13.9	187.9	15.0	14.2
19	12	6700	1.13	1.03	10.8	3.0	45.7	53.4	10.5	309.4	9.6	19.8
20	14	25000	1.17	1.03	13.4	3.4	80.6	47.7	6.1	251.4	41.0	17.3
21	12	7800	1.19	1.04	15.1	4.0	63.9	75.2	5.1	261.4	11.3	23.3
23	12	7500	1.13	1.02	10.8	1.8	132.3	58.3	3.7	333.2	29.3	16.2
24	12	17000	1.21	1.05	15.8	5.5	95.9	66.1	2.8	294.0	22.3	3.1
25	14	27000	1.14	1.03	11.1	2.8	47.2	75.1	6.1	235.0	17.7	18.6
27	12	23000	1.18	1.07	12.5	7.2	283.7	77.0	5.0	117.1	12.8	6.7
28	12	9200	1.12	1.02	9.8	2.5	351.6	82.3	4.8	187.8	5.8	18.3
29	12	16000	1.16	1.04	12.4	3.6	0.9	83.0	2.8	189.5	6.1	15.8
30	14	7700	1.15	1.02	12.3	2.4	4.2	84.8	3.4	175.0	4.1	15.3
31	12	18000	1.23	1.07	17.2	6.8	3.0	75.7	7.8	195.1	12.0	15.3

$\text{FOL} = 300 \times ((K_1 + K_2) \times 0.5) - K_3)/(K_1 + K_2 + K_3)$
$\text{LIN} = 300 \times (K_1 - K_2)/K_1 + K_2 + K_3$
a_{95} = Fisher's 95% confidence cone
n = number of specimens measured
* Errors in orientation suspected

Younger Dykes												
22	12	77000	1.05	1.02	3.6	1.8	87.4	1.0	15.8	182.5	22.5	13.8
26	12	150000	1.04	1.01	3.6	1.0	63.0	1.9	5.8	295.2	83.1	18.2
32	12	39000	1.03	1.02	2.1	2.3	23.7	56.8	11.9	210.5	37.0	10.8

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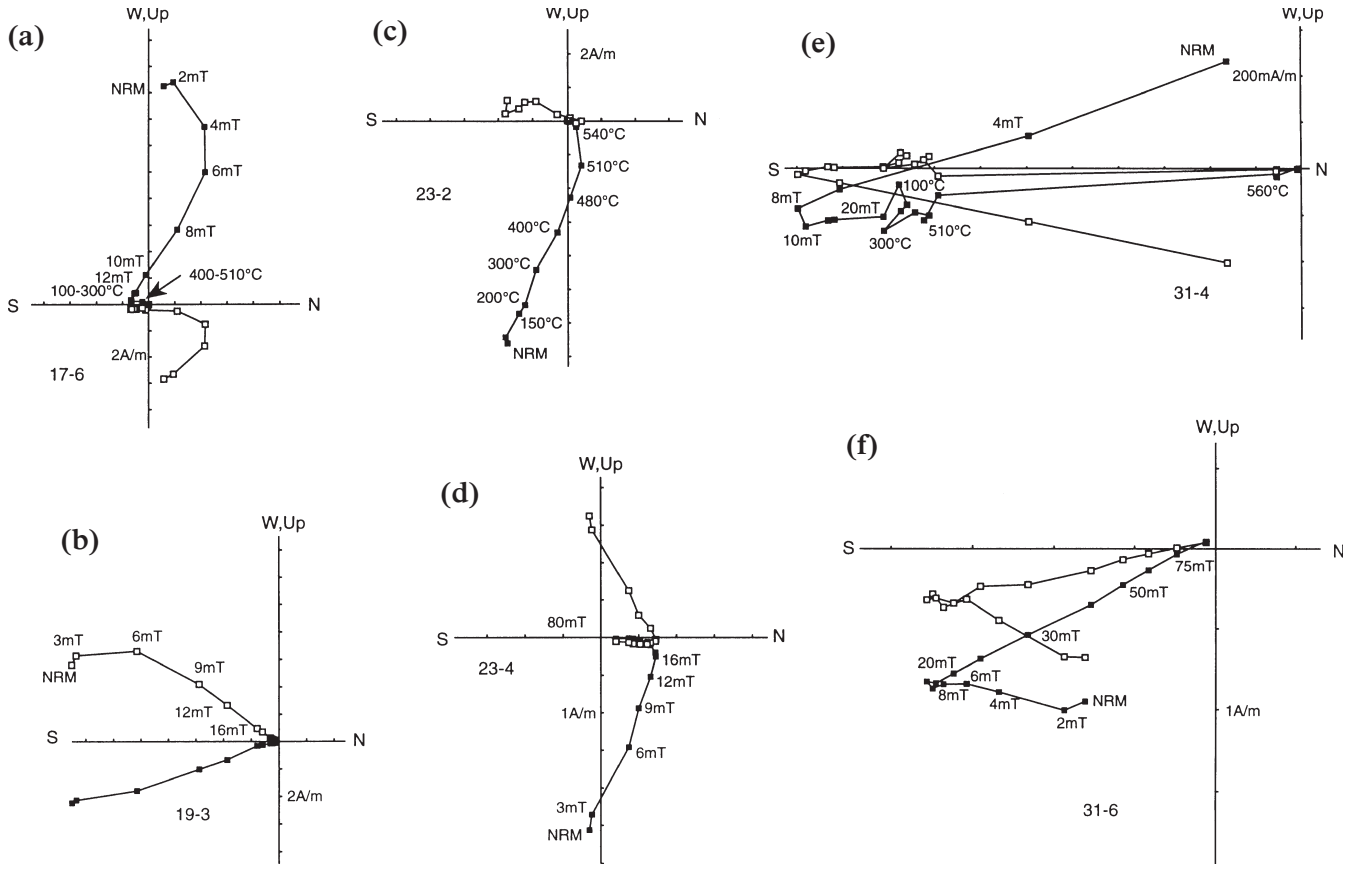


Figure 3. Representative Zijderveld diagrams showing behaviour on progressive demagnetization. (a) and (b) show a small reversed polarity ChRM revealed after AF demagnetization of large superimposed components; the ChRM in (a) has a high blocking temperature, and in (b) has a high coercivity of remanence. (c) and (d) show a small normal polarity ChRM after removal of large superimposed components by thermal and AF demagnetization, respectively, in a pair of samples from site 23. (e) and (f) illustrate that at site 31 the soft component and the ChRM are of comparable magnitude; in both cases the soft component removed in alternating fields of 10 mT is directed nearly horizontally to the North, whereas the ChRM has reversed polarity and high blocking temperature (e) and remanence coercivity (f).

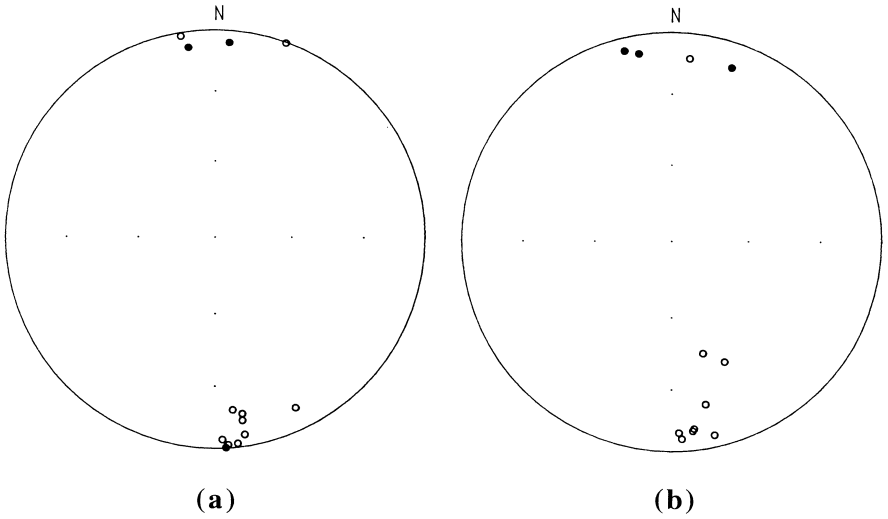


Figure 4. Stereographic projection of site-mean characteristic remanent magnetization directions (Table 1). (a) *in situ*. (b) tilt-corrected by rotating the magnetic foliation to the horizontal (site 18 excluded because no foliation was discernible).

MAGNETIC FABRIC

Anisotropy of magnetic susceptibility was measured in two specimens from all samples collected at the 17 sites (Table 2). Whereas the samples for the three younger dykes were nearly isotropic, those from the Complex rocks displayed a strong fabric. For each of these, the strike and dip of the magnetic foliation (plane perpendicular to the averaged K_3) at each site

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is shown in Fig. 5(a). These may be compared with the strike and dip of the layering as presented by Wells (1962; our Fig. 1a), and, as might be expected, with the exception of Site 27, the two generally coincide. The reason for the difference at Site 27 is not known. While the magnetic foliation (and lineation, see below) seems anomalous, the ChRM is not, and hence the magnetic fabric must be primary. Further study of this site is warranted.

Knowledge of whether the magnetic foliation acquired its present tilt before or after the characteristic remanence was acquired (i.e. above or below 570 °C) is relevant to the cooling and stress history of the Complex. To test this, the dispersions of ChRM *in situ* and relative to the foliation are compared. (Note that the calculation was based on results from 12 of the 13 sites where ChRM had been determined, because foliation was not observed at site 18.) The Fisher precision estimates, k, are 57.8 and 20, respectively. This contrast is sufficiently great (ratio 2.89) that it is sufficient to use the simple McElhinny (1964) fold test to assert that the ChRM was acquired after the foliation reached its present attitude, with confidence >0.95, because the critical ratio: k(*in situ*)/k(tilt-corrected) at $P = 0.05$ is 2.05.

As well as the foliation, a distinct magnetic lineation (direction of K_1 in the K_1–K_2 plane) is evident, and as shown in Fig. 5(b), it is consistently down-dip except at site 19, where it is almost parallel to the strike. Whereas the rocks sampled in this study ranged in general from troctolite, through gabbro to anorthosite, that at site 19 (an outcrop on shore at the Bawbaw beach house) was extremely olivine-rich. It is a dunite, and probably comes from the base of Wells' Zone 4 (see Plate 1 in Wells 1962).

The predominant down-dip orientation of the magnetic lineation, however, recalls the observation of Wells (1962; p. 36) of the presence of 'very dark pyroxene crystals' with an 'elongated tabular habit, the elongation being parallel to the direction of dip of the layering. They provide the only examples of linear orientation readily visible in the field.' Wells & Bowles (1981) elaborated on such pyroxene occurrences, and concluded from their study 'that the layered rocks of the Freetown Complex have been subjected to metamorphism. This occurred at near-magmatic temperatures during an extended cooling history under conditions of gradual deformation.' The fact that the palaeomagnetic fold test on our results is negative (see Fig. 4), however, demonstrates that the remanence was acquired with these rocks in their present position, after all structural adjustments had occurred.

In igneous rocks of this sort, the magnetic fabric is caused primarily by 'distribution anisotropy' (Hargraves *et al.* 1991; Stephenson 1994). In this, pre-existing silicates provide a form of template, controlling the distribution (and magnetic interaction) of late-crystallizing Fe–Ti oxides, whereby the AMS is a direct reflection of silicate fabric. At Freetown, the coincidence of the linear magnetic fabric with the 'metamorphic' pyroxene fabric described above is thus understandable, and the AMS demonstrates that this deformation was pervasive. The radial symmetry suggests that the stresses were related to late-stage adjustment during consolidation of the large, slowly cooled, quasi-cone-shaped body.

Pairs of petrographic thin sections were cut from three cores: one parallel to the foliation plane (K_1–K_2), and a second in a plane perpendicular to the foliation and parallel to K_1 (K_1–K_3). Whereas a preferred silicate grain orientation is obvious in the sections cut perpendicular to the foliation [tabular plagioclase and elongate ferromagnesians (Figs 2a and b)], there is no clear evidence of alignment in the foliation plane (K_1–K_2 sections, Figs 2c and d). The apparent long dimension of plagioclase grains was measured by image analysis of the three K_1–K_2 sections (photocopies taken with a microfiche projector), but no clear indication of correlation with the K_1 direction was obtained. It is suspected that, whereas AMS measures fabric

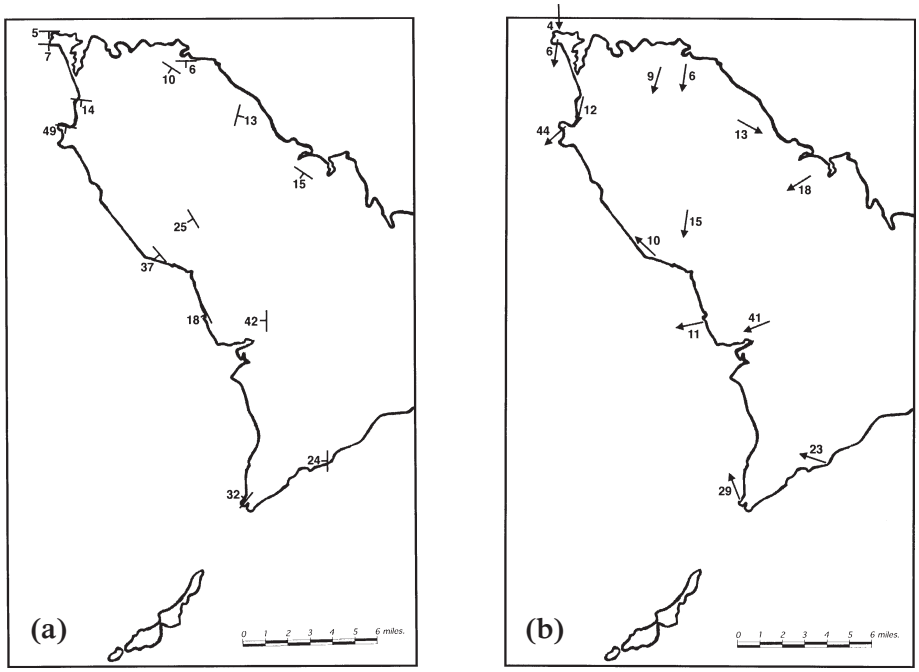


Figure 5. (a) Magnetic foliation from AMS; compare with layering in Fig. 1(a). (b) Magnetic lineation from AMS; compare with pyroxene lineation in Fig. 1(a).

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in the entire core, thin sections provide only a wafer sample through the core. When the rocks are as coarse-grained as these, and the linear anisotropy so subtle, petrographic means are inadequate to demonstrate the coincidence between K_1 and preferred silicate grain orientation.

The average bulk susceptibility values (x) of the samples at different sites are also shown in Table 2. The NRM intensities are highly variable by an order of magnitude or more, both within and between sites. Sample intensities range from 0.1 to 45 A m^{-1} due to the varying strength of IRM, which is almost certainly due to lightning, and Koenigsberger ratios calculated from these data would have little value.

The intensity of the ChRM [that remaining after the IRM and any viscous remanence (VRM) has been removed] was estimated, and found to be much more consistent among sites than the NRM. This ChRM is considered to be the best representation of the direction and provides an estimate of the minimum intensity of remanence for the buried bulk of the intrusion. ChRM values of 1 A m^{-1} or more are sufficiently strong that remanence should not be ignored when modelling airborne or shipborne magnetic surveys. The assertion of Mgbatogu *et al.* (1988) that N and R polarity zones within the intrusion influence their survey results is thus supported.

DISCUSSION

As reported above, the Freetown Complex palaeomagnetic pole obtained in the present study is indistinguishable from that reported by Briden *et al.* (1971) based on only 10 orientated hand samples. However, as a result of our more comprehensive sampling and laboratory study one can now confidently use the Freetown pole in palaeomagnetic pole comparisons. It is compared with other Jurassic poles from West Africa and the recently reported Karoo pole of Hargraves *et al.* (1997) in Table 3 and Fig. 6.

Published West African poles, except that from the Foum Zguid dyke, which may not represent a time-averaged palaeomagnetic field, fall in a group near 70°S, 60°E. The Freetown pole lies to the south of this group, and it is possible that it represents an older palaeomagnetic field. Although the ages of the published West African poles are not well constrained, it seems certain that they straddle the age of the Karoo (183 ± 2 Ma, Duncan *et al.* 1977), for which the pole lies at 69°S, 98°E. The possibility that this difference between the West African palaeomagnetic poles and that from the Kaapvaal craton reflects post-Jurassic intra(African)-plate deformation

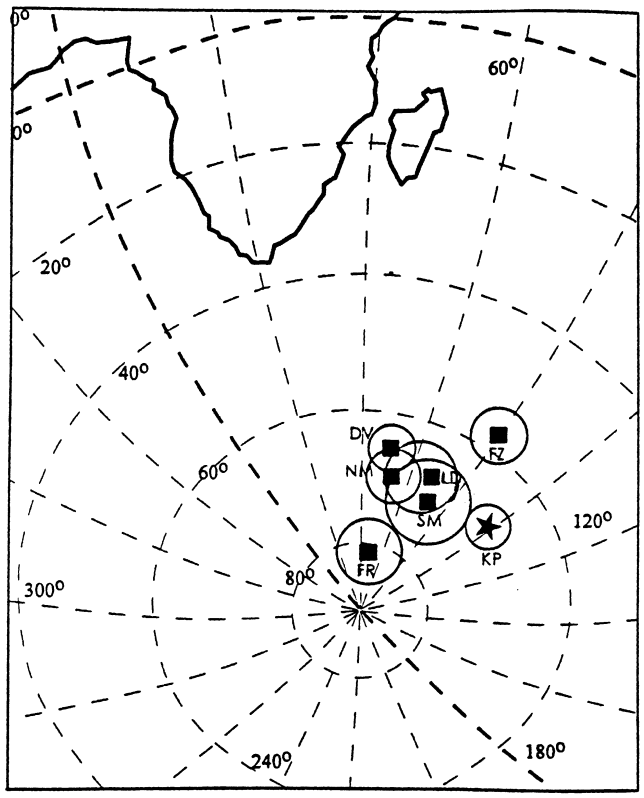


Figure 6. Comparison of West African Jurassic poles (solid squares, see Table 3) with the recently determined pole from Karoo igneous rocks in southern Africa (star).

akin to that inferred by Burke & Dewey (1974), Pindell & Dewey (1982), Rowley & Pindell (1989), has been suggested previously (Hargraves 1988; Kosterov & Perrin 1996; Hargraves *et al.* 1997). To this same end, Rapalini *et al.* (1993) averaged 20 late-Permian to Jurassic poles from West Africa (all published prior to 1981) and compared the result to an average calculated from nine (pre-1978) poles from southern Africa. Using the technique of McFadden & Lowes (1981) they compared the poles before and after the relative rotation model of Rowley & Pindell (1989). They concluded from this that 'the two populations have a common mean direction after correction for intracontinental deformation.' With improved dating and determination of palaeomagnetic poles the details of this deformation could be refined.

Table 3. West African and Karoo poles.

	Abbr.	No. of Sites	Lat°N	N. Pole Long°E	(*k*)*	α_{95}*	Probable Magnetization Age, Ma
Freetown Complex, S.L., This study	FR	13	82.9	212.7	56	5.6	193 ± 3 Rb-Sr
Foum Zguid dyke, Morocco [1]	FZ	5	58.0	259.0	(316)	(4.0)	182 – 187, K-Ar
Draa Valley sills, Morocco [1]	DV	16	65.5	230.5	(70)	(3.5)	180 – 186, K-Ar
North Mauretania dolerites [2]	NM	13	69.4	232	99	4.1	187 ± 5 K-Ar
South Mauretania dolerites [2]	SM	13	71.4	240.2	47	6.1	187 ± 5 K-Ar
Liberian dykes [3]	LD	25	68.5	242.4	31	5.3	173 – 192 K-Ar
La Reculée sediments [4]	LR	8	70.9	235.2	478	2.3	180 – 210 strat.
Karoo Igneous Province [5]	KP	6 localities	69.2	278.3	416	3.3	183 ± 2, Ar-Ar

* Brackets around k and α_{95} signify statistics provided for mean study directions, not mean of poles. References: [1] Hailwood & Mitchell (1971); [2] Sichler *et al.* (1980); [3] Dalrymple *et al.* (1975); [4] Kies *et al.* (1995); [5] Hargraves *et al.* (1997).

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While congruence between the magnetic foliation and the layering could be expected, that between the magnetic lineation and the macroscopic pyroxene fabric reported by previous workers (Bowles 1977; Wells & Bowles 1981) is noteworthy. The pyroxene fabric, however, seems to have been observed in the field at only a few sites [see Fig. 1a, based on Wells's (1962) map], whereas the magnetic lineation is pervasive. This testifies to the utility of AMS in studies addressing fluid flow and crystal growth and accumulation—and possibly deformation—in such igneous plutons generally.

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